Impacts and Trends of Computing Technology

WI1102 Computational Thinking Teaching Team

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Objectives

- Students are able to explain the impacts and trends of computing technology from various perspectives, particularly those relevant to their own field of study.

The Digital World

- Technological advancements, particularly in the field of computing, have brought about changes in how we live.
- Computers have become an essential tool in any profession; PCs—including laptops—are no longer merely office equipment, but also serve as devices for entertainment.
- The realm of the Internet—cyberspace—is a *new* world that anyone can enter, unconstrained by time or space. It is human beings who step into this virtual world.

The Digital World as a Source for Lifelong Learning

- The digital world has become an inexhaustible source of learning. Educational materials are readily available and easily accessible.
- Reference resources—such as encyclopedias, dictionaries, atlases, and medical references—are accessible in digital formats via the Internet, smartphone applications, and CD/DVDs.
- The Internet offers blogs, forums, websites, and a wide array of media for learning anything, anywhere, thereby facilitating distance learning.
- Edutainment combines education with entertainment.

Computing technology also has an impact on the workplace

- Computing technology brings efficiency to various human tasks.
- Automation, driven by information technology (IT), unlocks a vast array of new job opportunities.
- Certain technologies also “unlock” human creativity.
  - AI (Artificial Intelligence) ??
- Globalization: creates a “worldwide workforce” and opens up markets on a global scale.
- At the same time: the use of computing technology may threaten jobs replaceable by machines

Discussions

- Name examples of digital technology (computers, the internet, gadgets, etc.) that you find highly supportive of your activities as a student (studying, organizational involvement, etc.).

- Imagine if you did not have access to these technologies—how would you go about your activities?

Computing Technology in Daily Lives

Everyday life already depends on computing technology…

- Many aspects of daily life have become highly dependent on computing technology:
  - Household/family needs: online shopping, delivery services, etc.
  - Entertainment: movies, games, video streaming, etc.
  - Transportation: booking taxi services, airline tickets, train tickets, etc.

  Identify and explain other examples.
- Some potential threats:
  - The risks of "addiction" and "violence"
  - Identity theft
  - Threats to privacy

OCT 2025

OVERVIEW OF SOCIAL MEDIA USE

HEADLINES FOR SOCIAL MEDIA ADOPTION AND USE (NOTE: USER IDENTITIES MAY NOT REPRESENT UNIQUE INDIVIDUALS)



INDONESIA

NUMBER OF SOCIAL MEDIA USER IDENTITIES



180

MILLION

QUARTER-ON-QUARTER CHANGE IN SOCIAL MEDIA USER IDENTITIES



-7.3%

-14 MILLION

YEAR-ON-YEAR CHANGE IN SOCIAL MEDIA USER IDENTITIES



+26.0%

+37 MILLION

WEEKLY TIME SPENT USING SOCIAL PLATFORMS (INC. VIDEO)



21H 50M

AVERAGE NUMBER OF SOCIAL PLATFORMS USED EACH MONTH



7.7

YOY: [UNCHANGED]

SOCIAL MEDIA USER IDENTITIES vs. TOTAL POPULATION



62.9%

SOCIAL MEDIA USER IDENTITIES AGED 18+ vs. POPULATION AGED 18+



88.9%

SOCIAL MEDIA USER IDENTITIES vs. INDIVIDUALS USING THE INTERNET



78.2%

FEMALE SOCIAL MEDIA USER IDENTITIES vs. TOTAL SOCIAL MEDIA USER IDENTITIES



56.3%

MALE SOCIAL MEDIA USER IDENTITIES vs. TOTAL SOCIAL MEDIA USER IDENTITIES



43.7%

SOURCES: KEPIOS ANALYSIS; COMPANY ADVERTISING RESOURCES; CNNIC; BETA RESEARCH CENTER; OCDH; U.N.; GWI (Q2 2025). **NOTES:** WHERE PRESENT, "[BASE REVISIONS]" INDICATES THAT VALUES FOR TRENDS OVER TIME ARE NOT REPRESENTATIVE OF CHANGES IN ACTUAL SOCIAL MEDIA USE. AVERAGE NUMBER OF PLATFORMS INCLUDES DATA FOR YOUTUBE. **ADVISORY:** SOCIAL MEDIA USER IDENTITIES MAY NOT REPRESENT UNIQUE INDIVIDUALS. COMPARISONS WITH POPULATION AND INTERNET USERS MAY SHOW MORE THAN 100% DUE TO MULTIPLE ACCOUNTS, USER AGE MISSTATEMENTS, DIFFERENT REPORTING PERIODS, AND DIFFERENCES BETWEEN CENSUS COUNTS AND RESIDENT POPULATIONS. **COMPARABILITY:** SOURCE AND METHODOLOGY CHANGES; BASE REVISIONS. SEE NOTES ON DATA.

WI1102/Computational Thinking

we are social

Meltwater

OCT 2025

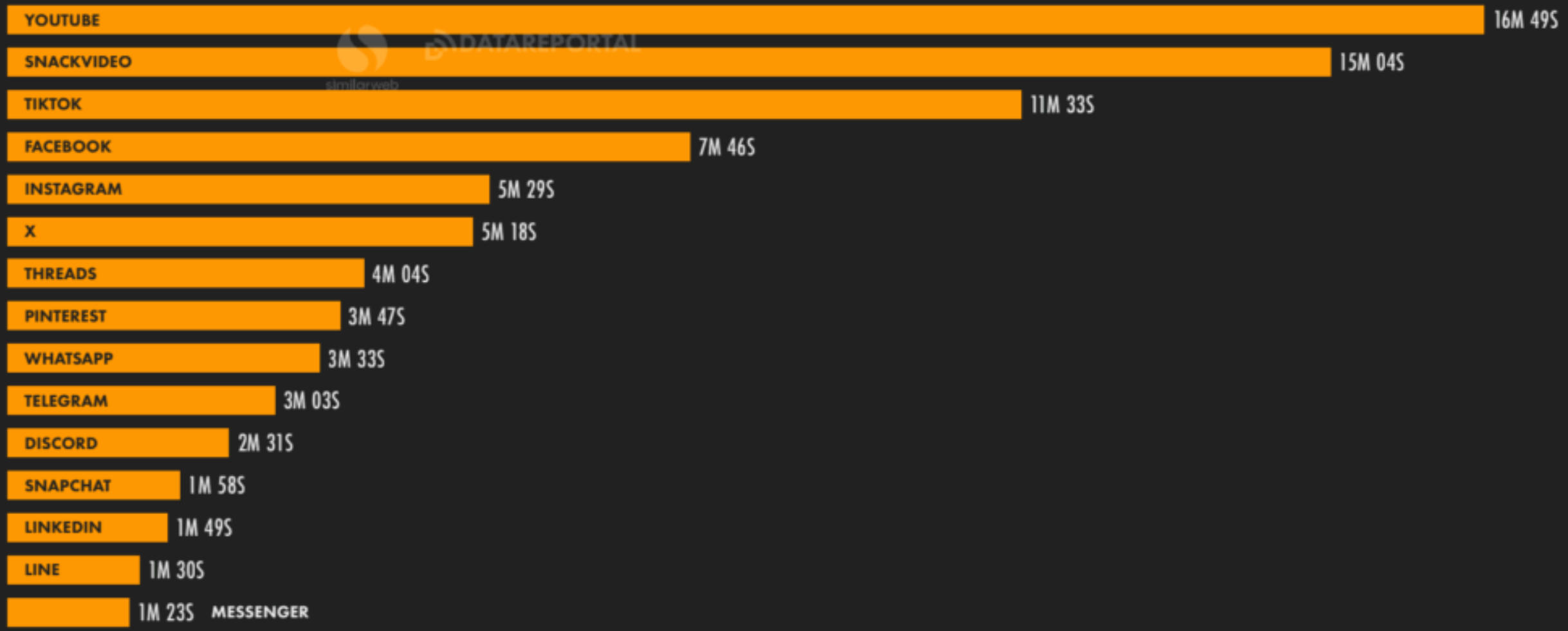
SOCIAL MEDIA APPS: AVERAGE SESSION DURATION

AVERAGE SESSION DURATION (IN MINUTES AND SECONDS) EACH TIME A USER OPENS THE RESPECTIVE PLATFORM'S ANDROID APP

NOTE: THE DATA SOURCE THAT INFORMS THIS CHART HAS CHANGED, SO VALUES ARE NOT COMPARABLE WITH PREVIOUS REPORTS



INDONESIA



DATAREPORTAL
similarweb

SOURCE: SIMILARWEB APP INTELLIGENCE. NOTE: ONLY INCLUDES DATA FOR THE TOP 15 SOCIAL MEDIA APPS RANKED BY AVERAGE MONTHLY ACTIVE USERS. FIGURES REPRESENT THE AVERAGE AMOUNT OF TIME (IN MINUTES AND SECONDS) THAT USERS SPENT USING THE RESPECTIVE PLATFORM'S ANDROID APP(S) EACH TIME THEY OPENED THE RELEVANT APP(S) IN AUGUST 2025. WHERE APPLICABLE, VALUES REPRESENT AGGREGATES FOR MULTIPLE APPS (E.G. PRIMARY AND "LITE" VERSIONS). COMPARABILITY: SOURCE CHANGE. VALUES ARE NOT COMPARABLE WITH PREVIOUS REPORTS.

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we are social

Meltwater

Threats of Violence via the Internet

- **Cyberbullying**
  - Cyberbullying is a form of verbal or psychological abuse carried out through digital platforms, such as social media, instant messaging, or online forums.
- **Grooming and Exploitation**
  - Grooming is an attempt by a predator to build a relationship with a young or vulnerable individual via technology for the purpose of exploitation or abuse.
- **Trolling, cyberstalking, doxxing**



Threats of Violence via the Internet

- **The Spread of Visual Violence and Extreme Content**
  - The distribution of violent content via the internet, such as videos of violence, extremism, or radical propaganda.
  - **Cyberviolence** is *the use of computer systems to cause, facilitate, or threaten violence against individuals that results in, or is likely to result in, physical, sexual, psychological or economic harm or suffering and may include the exploitation of the individual's circumstances, characteristics or vulnerabilities.* https://www.coe.int/en/web/cyberviolence/cyberviolence-at-a-glance



Internet Addiction

- **Internet addiction** is the excessive use of the internet to the point that it interferes with one's social life, work, and education.
- Adolescents are an age group particularly vulnerable to internet addiction; with up to 31.4% of adolescents in Indonesia suffer from it (2021 data).

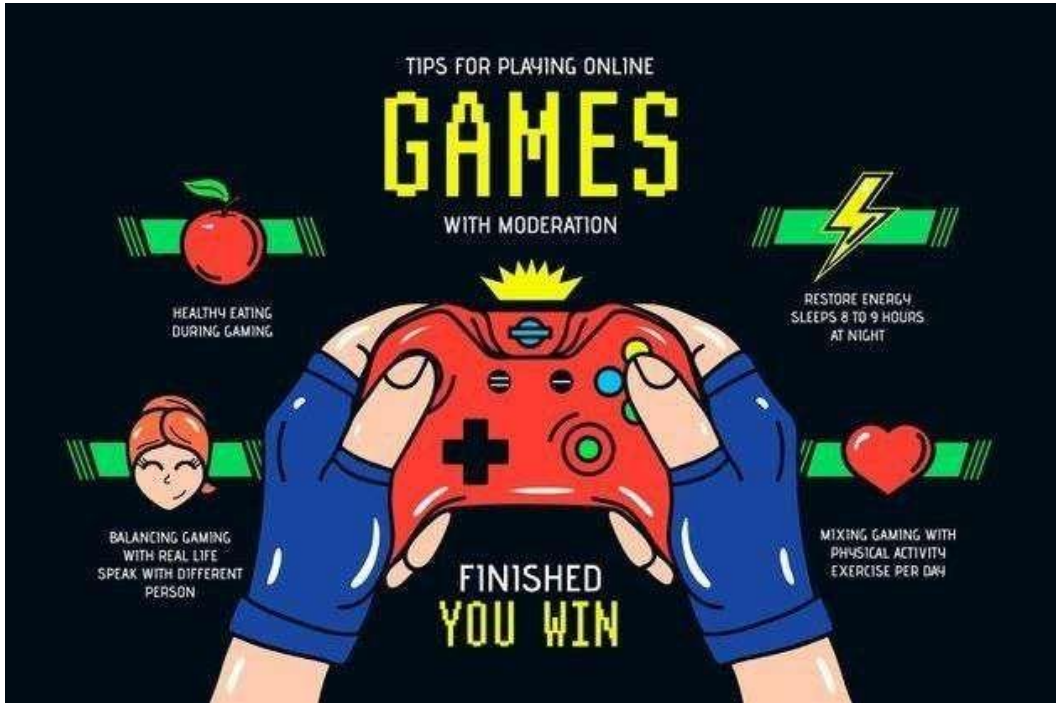


https://www.cnnindonesia.com/gaya-hidup/20211002135419-255-702502/survei-193-persen-anak-indonesia-kecanduan-internet

- Some of the consequences include: a decline in interest in learning, mental and behavioral changes, emotional instability, hallucinations, and even severe mental disorders.

Gaming Addiction

- Gaming can be beneficial: it can reduce stress, enhance concentration and creativity, improve decision-making skills, and even generate income.
- **Gaming addiction**: an addiction to video games or online games that interferes with daily activities, work, and social relationships.
- According to a 2023 survey by the Indonesian Internet Service Providers Association (APJII), 42.23% of online gamers spend more than 4 hours a day playing online games.



https://www.youtube.com/watch?v=BicDbsfxtfA

Avoiding Addiction & Violence

- Pursue hobbies and interests unrelated to the internet; participate in social activities such as student organizations, associations, or clubs.
- Engage in more physical activities, such as play, sports, or recreation.
- Socialize with people in the real world rather than virtual friends.
- If you encounter a problem, discuss it in person.
- Maintain good relationships with family and friends.
- If you develop an addiction, talk about it with your parents, a teacher, or a doctor.
  - Avoid trying to find all the answers on your own via the internet.
- Limit the time spent using the internet on laptops and other gadgets.
- Delete unnecessary applications.
- Use filtering software (especially for children).

Discussions

- **How many hours** a day do you spend on the following:
  - Accessing the internet
  - Playing online games
- What are the **positive** aspects of your daily habits regarding internet use and gaming?
- What are the **negative** aspects of your daily habits regarding internet use and gaming?

Personal Data: All about You

- **Identity theft** has become a substantial criminal industry.
  - It can occur due to carelessness while online or a simple miscalculation.
  - Consequences include receiving large, unauthorized phone or internet bills, being threatened by strangers, or having one's social media accounts used to spread slander or issue threats—which could even lead to arrest by the authorities, among other issues.
  - Resolving the repercussions of identity theft can take a very long time—potentially even years.



Protecting Yourself from Identity Theft

- Make your online purchases using a reputable credit card or electronic payment.
- Scan your bills promptly.
- Get a separate credit card for online transactions.
- Make sure a secure Web site is managing your online transaction.
- Don’t disclose personal information over the phone or chats.
- Handle email and social media with care.
- Don’t put your Social Security number on your checks.
- Shred sensitive mail before you recycle it.
- Keep your wallet thin.
- Copy your cards.
- Report identify theft promptly.

Threats to Privacy

- Placing personal data online means it gets stored in a server database.
  - It can spread very easily across the internet.
  - Be aware of your rights regarding your data.
- E-commerce companies may use personal data for various purposes (such as advertising).
- Privacy is increasingly diminishing these days.
  - In the workplace, strict monitoring of employees can be carried out using computing technology.
  - Surveillance cameras are used in public spaces.
  - Information technology makes it possible to conduct surveillance or wiretapping on individuals, organizations, or even entire nations.

Your Privacy Rights

- Do not readily share or distribute your national ID number (KTP), driver's license number, phone number, email address, mobile number, bank account number, or other sensitive data—whether your own or someone else's.
- Do not readily provide personal information.
- Do not readily consent to the sharing of personal information.
- THINK before you POST!
- Use technology to protect your personal rights.
- If there is damaging information about you, correct it!

Pasal 1

Dalam Undang-Undang ini yang dimaksud dengan:

1. Data Pribadi adalah data tentang orang perseorangan yang teridentifikasi atau dapat diidentifikasi secara tersendiri atau dikombinasi dengan informasi lainnya baik secara langsung maupun tidak langsung melalui sistem elektronik atau nonelektronik.
2. Pelindungan Data Pribadi adalah keseluruhan upaya untuk melindungi Data Pribadi dalam rangkaian pemrosesan Data Pribadi guna menjamin hak konstitusional subjek Data Pribadi.

Pasal 65

(1) Setiap Orang dilarang secara melawan hukum memperoleh atau mengumpulkan Data Pribadi yang bukan miliknya dengan maksud untuk menguntungkan diri sendiri atau orang lain yang dapat mengakibatkan kerugian Subjek Data Pribadi.

(2) Setiap Orang dilarang secara melawan hukum mengungkapkan Data Pribadi yang bukan miliknya.

(3) Setiap Orang dilarang secara melawan hukum menggunakan Data Pribadi yang bukan miliknya.

Law No. 27 of 2022 concerning Personal Data Protection

Discussions

- Look for the Privacy Policy of the social media application(s) you use.
- How is your personal data used by that social media application?
- Do you consider this acceptable?

Computing Technology in Ethics, Society, Politics, and Economics

Norms, Ethics, and Regulations

- **Norms** and **ethics**: relate to morality and serve merely as "guidelines"; your own conscience must play a role.
- **Rules** and **regulations**: these are more operational in nature, explicitly stating requirements, regulations, penalties, and sanctions. Rules are established to make communal life "comfortable," not to create difficulties.
- Violating norms or ethics should evoke a sense of "shame." Violating rules, however, results in punishment—a measure intended to ensure that members of the community are aware of the rules and refrain from breaking them.
- Look for resources on the Internet using a search engine by entering keywords such as "**netiquette**" or "**net-ethics**."

Computer Ethics

- Know the rules and the law.
- Don’t assume that it’s okay if it’s legal.
- Think scenarios.
- When in doubt, talk it out.
- Make yourself proud.
- Remember the golden rule.
- Take the long view.
- Do your part.



https://infocompethics.game.blog/2018/08/20/why-is-it-important-to-follow-computer-ethics/

Netiquette and Messaging Etiquette

- Say what you mean; say it with care.
- Keep it short.
- Proofread your messages.
- Don’t assume you are anonymous.
- Learn the “nonverbal” language of the Net.
- Know your abbreviations.
- Keep your cool.
- Don’t be a source of spam.
- Say no—and say nothing—to spam.

Freedom vs. Social Responsibility

- The internet provides the freedom to speak and express opinions, which can also be exploited for irresponsible purposes.
- The government has established regulations: **Law No. 11/2008 on Information and Electronic Transactions**.
  - Law No. 19 of 2016 concerning Amendments to Law No. 11 of 2008 on Information and Electronic Transactions.
  - Law No. 1 of 2024 concerning the Second Amendment to Law No. 11 of 2008 on Information and Electronic Transactions.
- We remain responsible for whatever we think, say, or write, whether online or offline.
- Ethics are moral principles or standards that serve as a guide for behavior, actions, and choices.

Social Networking

- WhatsApp, Instagram, Facebook, X (formerly Twitter), … to name a few.
- They connect people across the globe, creating a world without the constraints of space or time.
- It is crucial to be fully aware of both their benefits and their drawbacks!
- While they can serve as tools for work-related communication, exercise caution regarding the information you share. Avoid disseminating information intended only for specific individuals or groups to everyone—make use of group features instead.

Social Networking

- Be careful when writing status updates or tweets! Think before you hit "send"! Your posts can have an impact all around the world.
- Do not share content—whether text, images, videos, or otherwise—that involves SARA (issues related to ethnicity, religion, race, or societal groups), pornography, illegal material, or anything else that could cause public unrest. There has already been a case where an ITB student was suspended for sharing content related to SARA issues.
- Limit the amount of personal information you share in the public sphere.
  - Video: https://www.youtube.com/watch?v=M8dzxgYwhrI

Be careful on what you post online!

Pasal 27

(3) Setiap Orang dengan sengaja dan tanpa hak mendistribusikan dan/atau mentransmisikan dan/atau membuat dapat diaksesnya Informasi Elektronik dan/atau Dokumen Elektronik yang memiliki **muatan penghinaan dan/atau pencemaran nama baik**.

Pasal 28

(1) Setiap Orang dengan sengaja dan tanpa hak menyebarkan **berita bohong dan menyesatkan** yang mengakibatkan kerugian konsumen dalam Transaksi Elektronik.

(2) Setiap Orang dengan sengaja dan tanpa hak menyebarkan informasi yang ditujukan untuk **menimbulkan rasa kebencian atau permusuhan individu dan/atau kelompok masyarakat tertentu berdasarkan atas suku, agama, ras, dan antargolongan (SARA)**.

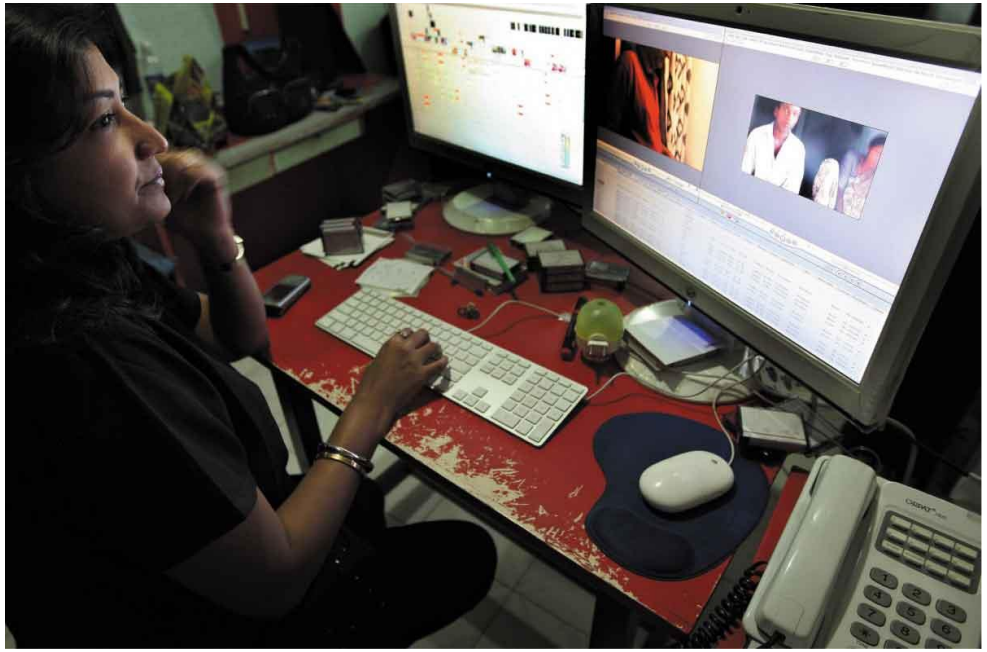
Pasal 45

(1) Setiap Orang yang memenuhi unsur sebagaimana dimaksud dalam Pasal 27 ayat (1), ayat (2), ayat (3), atau ayat (4) dipidana dengan pidana penjara paling lama 6 (enam) tahun dan/atau denda paling banyak Rp1.000.000.000,00 (satu miliar rupiah).

(2) Setiap Orang yang memenuhi unsur sebagaimana dimaksud dalam Pasal 28 ayat (1) atau ayat (2) dipidana dengan pidana penjara paling lama 6 (enam) tahun dan/atau denda paling banyak Rp1.000.000.000,00 (satu miliar rupiah).

Where Computers Work

- Many jobs have been transformed by digital technology:
  - Entertainment
  - Publishing
  - Medicine
  - Airlines
  - Science



The Automated Factory

- **Robots**: used to perform specific manual tasks
  - Painting, welding, and repetitive assembly-line jobs
- Computers track inventory, time delivery of parts, control the quality of production, monitor wear and tear on machines, and schedule maintenance.
- Automation reduces the time that materials and machines sit idle and reduces waste in facilities, raw materials, and labor.

The Automated Office

- As factory jobs decline, office work plays a more important role in our economy.
- Modern offices, like modern factories, have been transformed by digital technology.



Automation, Globalization, and Outsourcing

- Many analysts predicted that automation would lead to massive unemployment and economic disaster.
- Others said that computers would generate countless new job opportunities.
- Today, most people are accustomed to seeing computers where they work.
- Job automation is still an important issue for millions of workers whose jobs are threatened by machines.

Discussions

- The current job market is increasingly global. Your competitors are not only from within the country but also from abroad.
- Do you view this as a threat, a challenge, or an opportunity?
- What is your approach to dealing with it?

Discussions

- The development of digital technology has accelerated to such an extent that many human tasks can now be performed by machines—including AI—with far greater efficiency.
- In your view, how should humans approach this phenomenon? Should it be rejected because it threatens human employment, or should it be embraced as an inevitable part of the changing times? What steps should be taken?

Artificial Intelligence (AI)

What is Artificial Intelligence (AI)



https://www.youtube.com/watch?v=4jmsHaJ7xEA

So.. What is AI? 8 Definitions, 4 Approaches

Thinking humanly

Thinking rationally

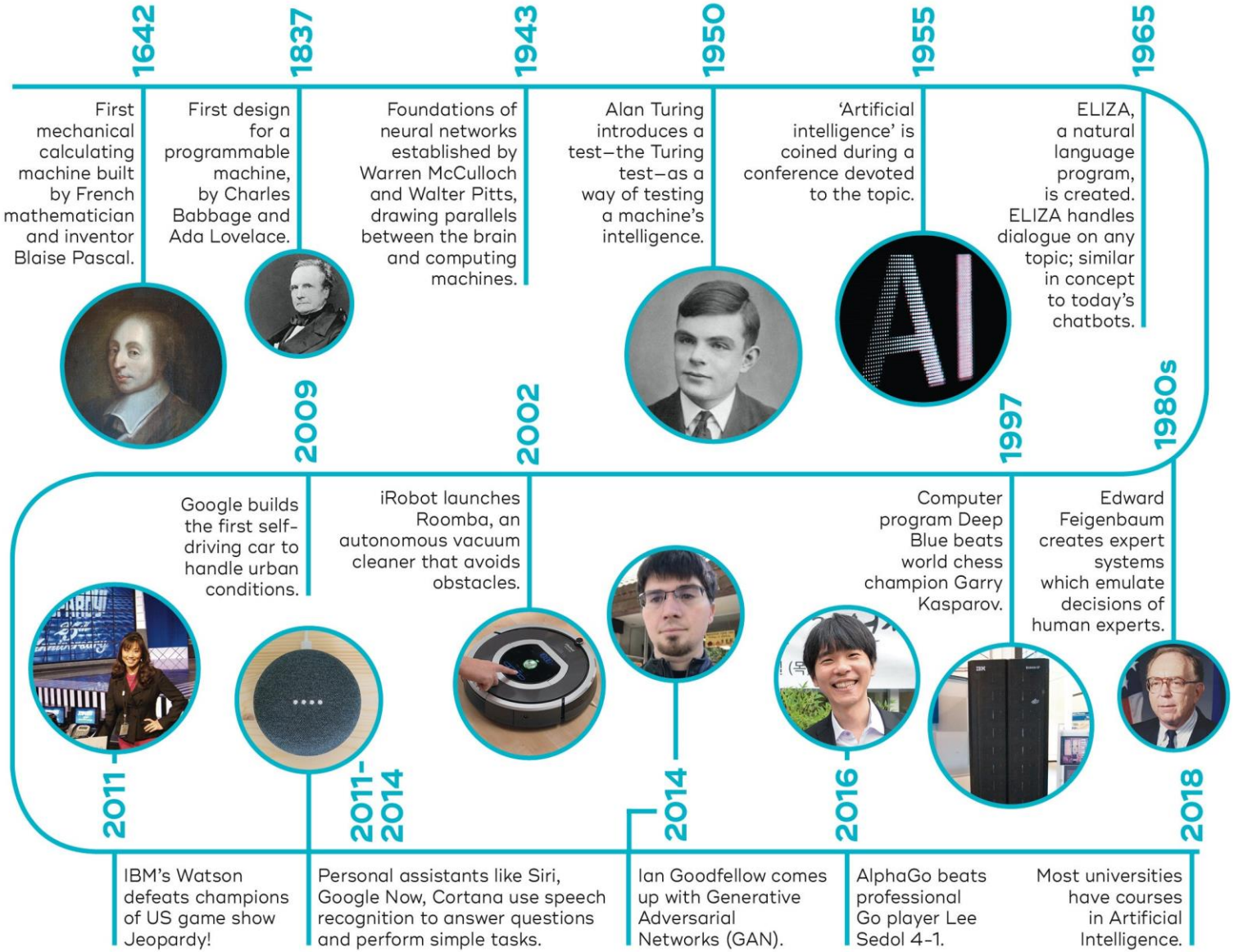
Acting humanly

Acting rationally

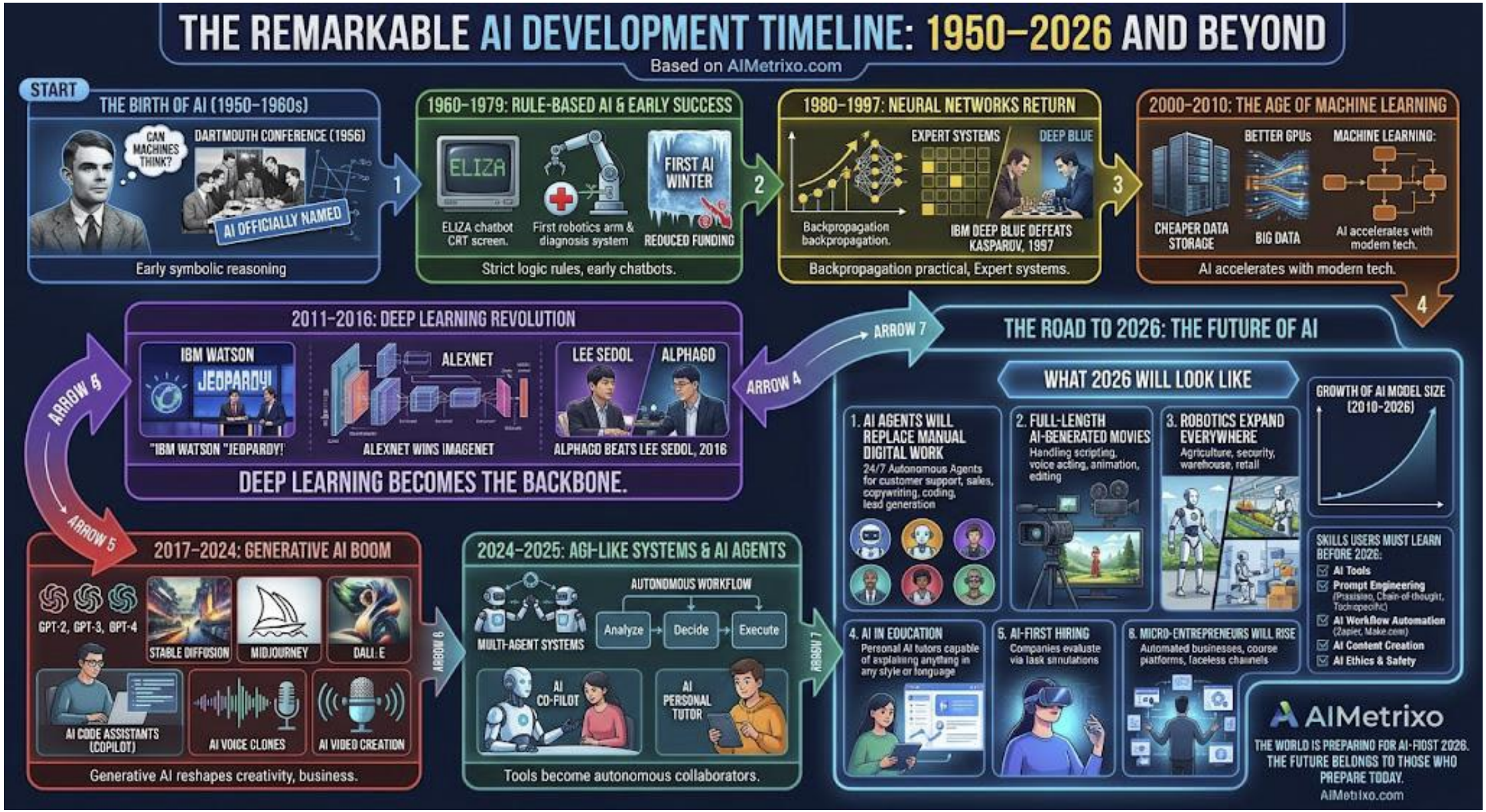
Thinking Humanly	**Thinking Rationally**
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Acting Humanly	**Acting Rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998) "AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

History of AI



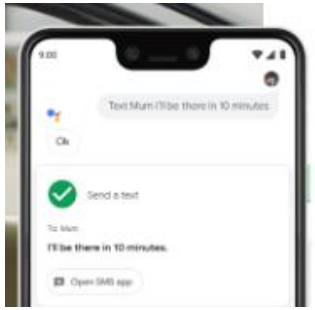
https://qbi.uq.edu.au/brain/intelligent-machines/history-artificial-intelligence



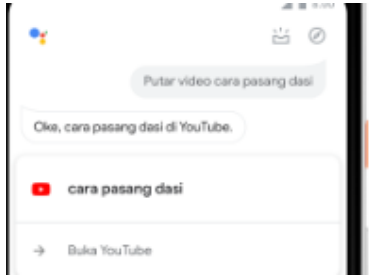
This image is generated by Google Gemini based on the contents of https://aimetrixo.com/ai-development-timeline/

AI In Life: AI Assistant





"Hey Google, text Mum I'll be there in 10 minutes"



"Hey Google, play my morning playlist"



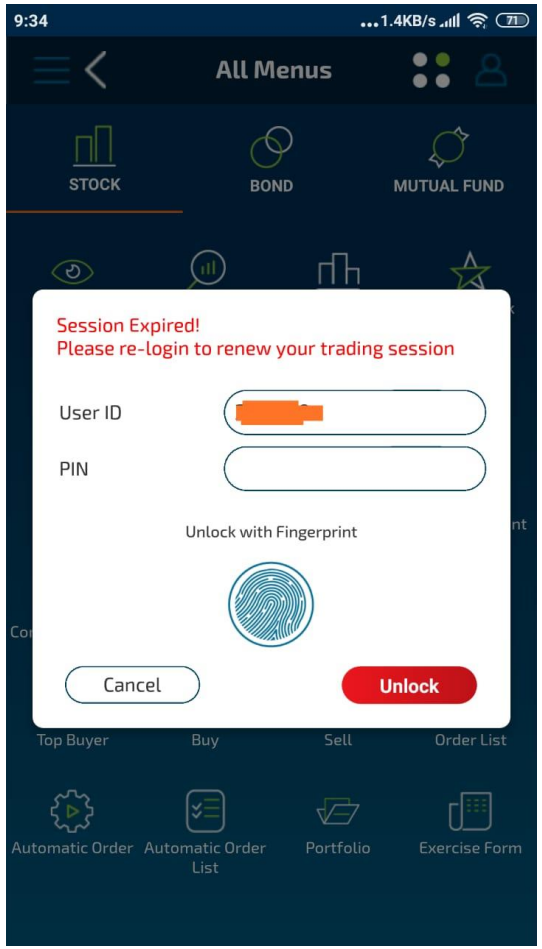
"Hey Google, dim the bedroom lights"



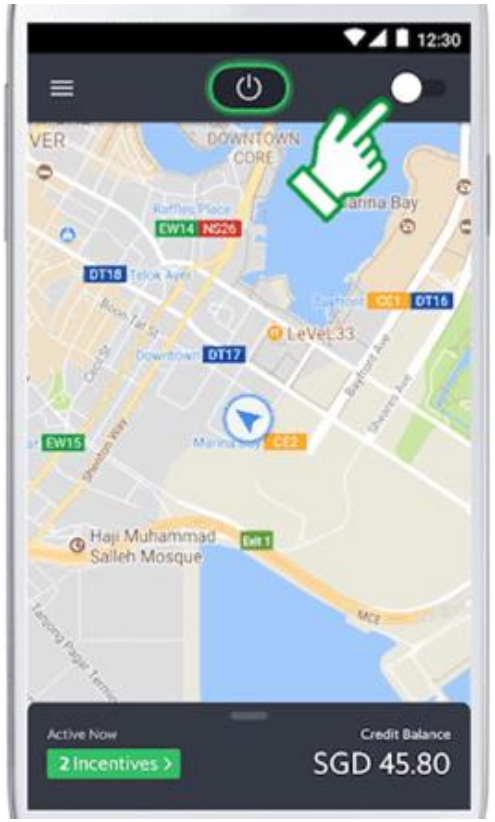
"Hey Google, set the temperature to 20 degrees"

AI In Life: Biometric Verification

Fingerprint verification



Face verification



https://www.grab.com/sg/driver-2/selfieverification/

AI In Life: OpenAI untuk Dota 2



OpenAI Five

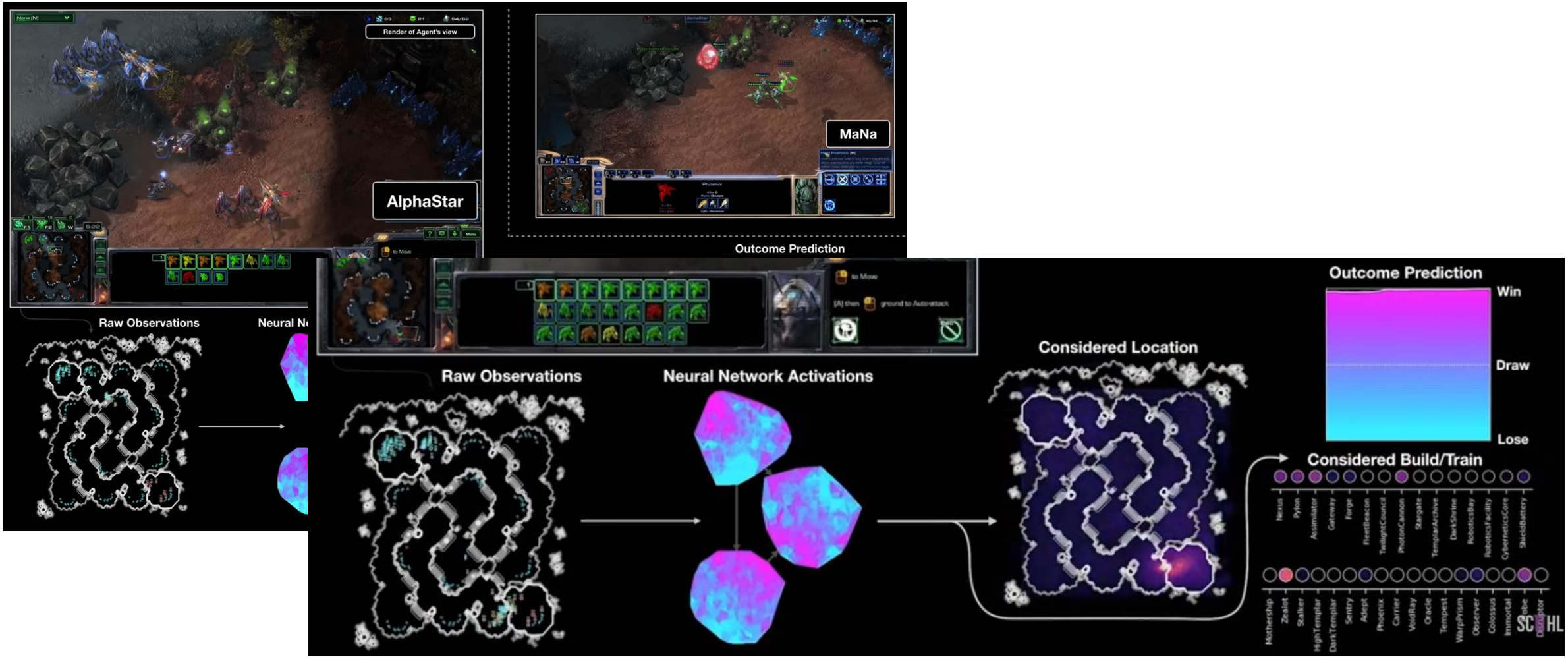
Competitive: 7.215–42 (99.4% winrate, 15.019 total players)

Note: During the live stream, the game count incorrectly omitted games abandoned by the human side.

Cooperative: 35.466 games (18.689 total players)

Source: https://i.ytimg.com/vi/6-UuJb8L8X8/maxresdefault.jpg

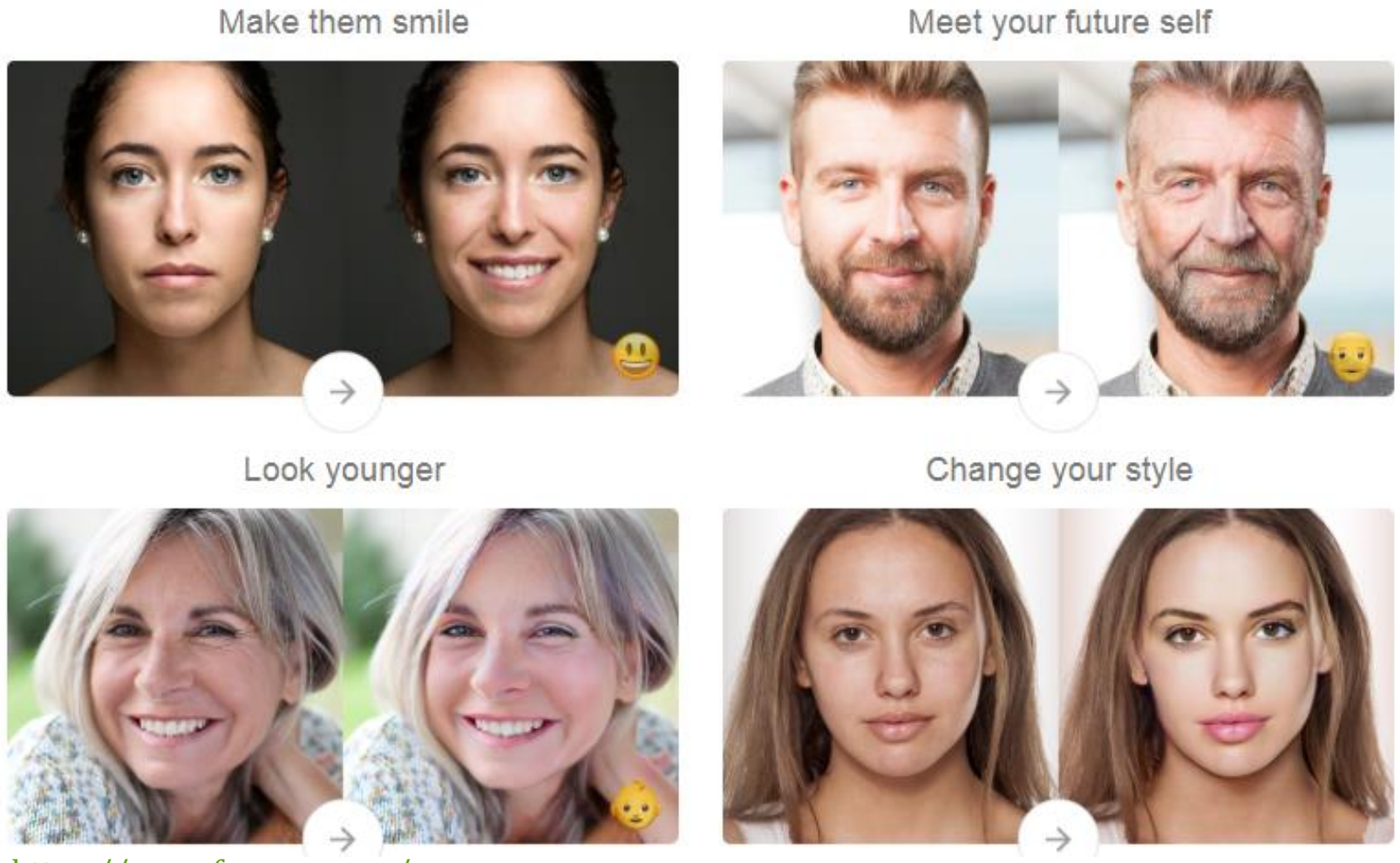
AI In Life: AI Alphastar StarCraft II



https://techcrunch.com/2019/01/24/starcraft-ii-playing-ai-alphastar-takes-out-pros-undefeated/

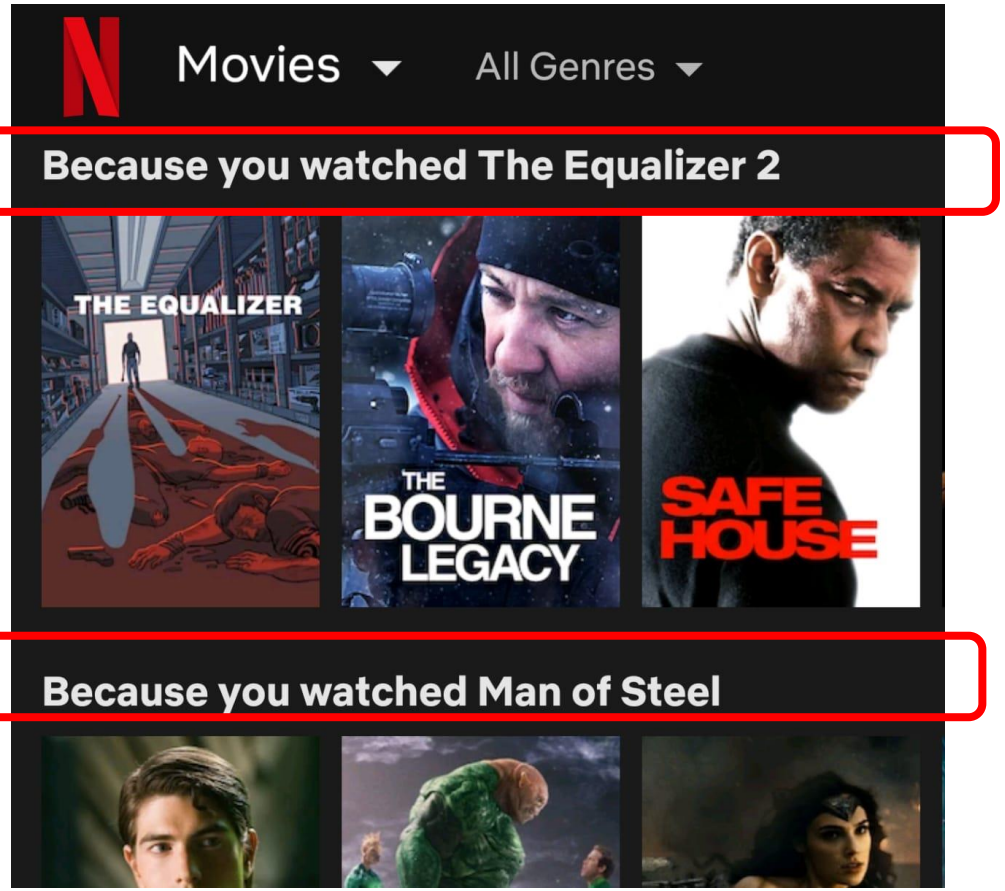
https://www.youtube.com/watch?v=PFMRDm_H9Sg

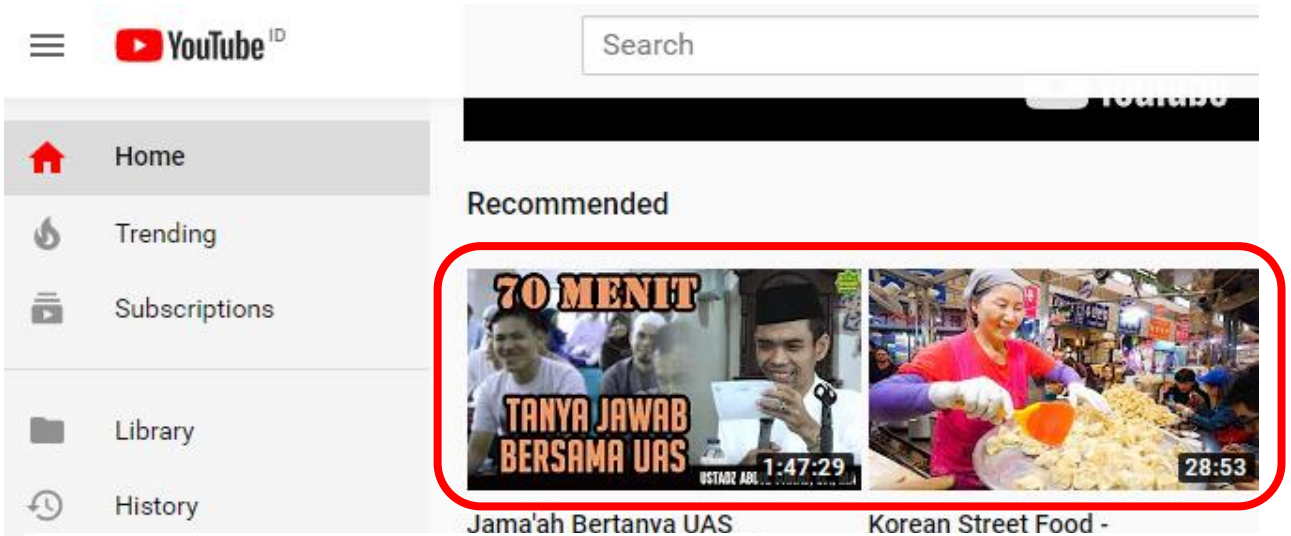
AI In Life: Transform Your Face

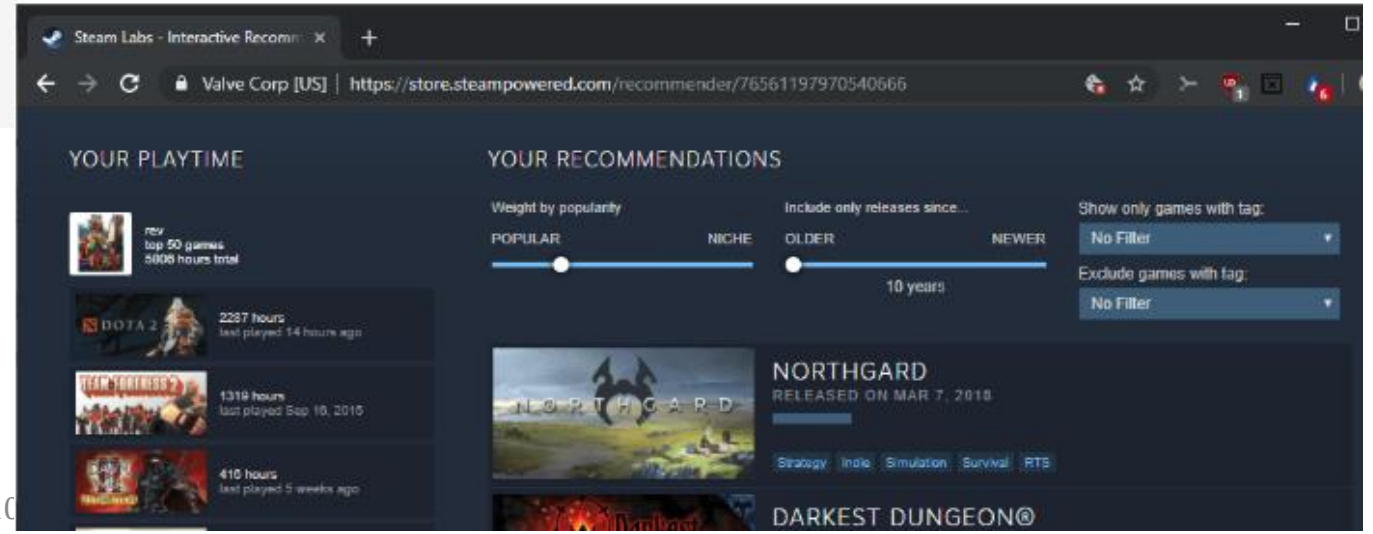


https://www.faceapp.com/

AI In Life: Recommender System





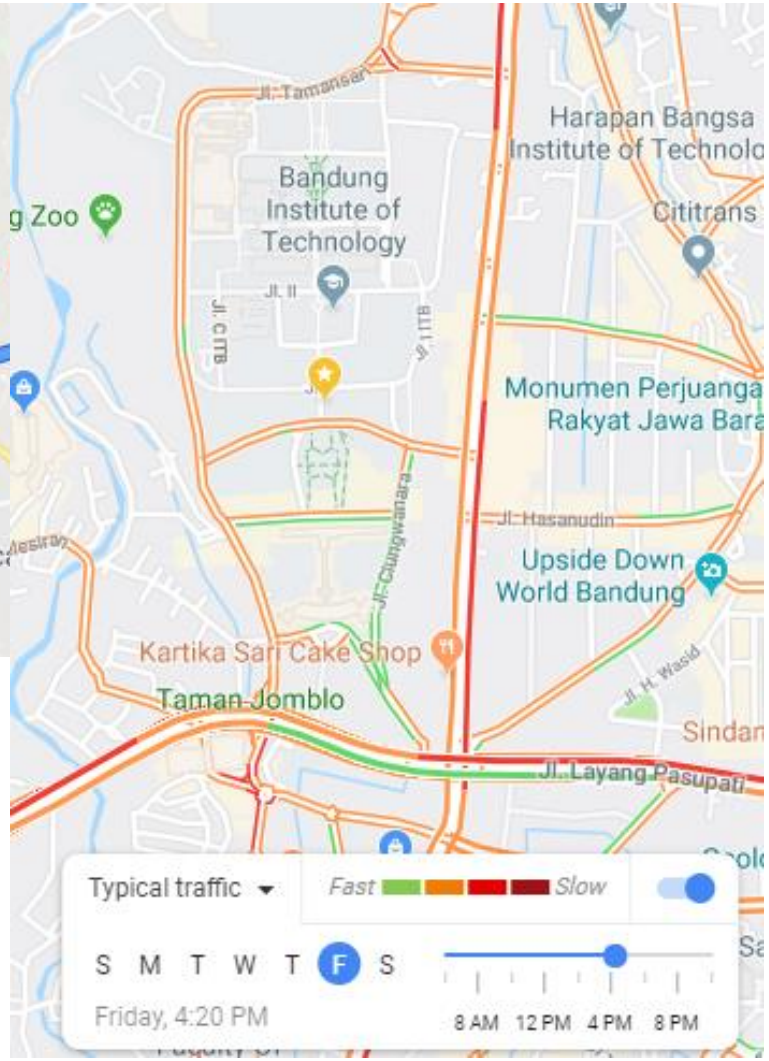




AI In Life: AI in Traffic



Path Finding / Direction



Traffic pattern

AI In Life: Fraud Detection



- Transactions made by credit cards in September 2013 by European cardholders.
- Deep Learning Approach: Autoencoder is a neural network that is trained to attempt to copy its input to its output.
- Risk Analytics to prevent fraud.

Source: https://miro.medium.com/max/3848/1*3R2cLLJ349O5G6V5rZJXxw.png

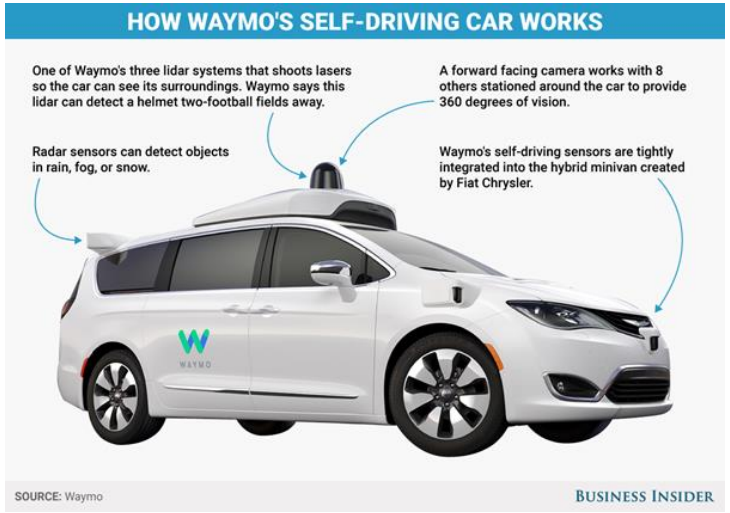
Self Driving Cars



Q2 2018: paid Singapore self-driving car rides
https://www.reuters.com/article/us-nutonomy-singapore-idUSKCN1AY2IC



https://www.techradar.com/news/uber-self-driving-cars
(May 25, 2018)

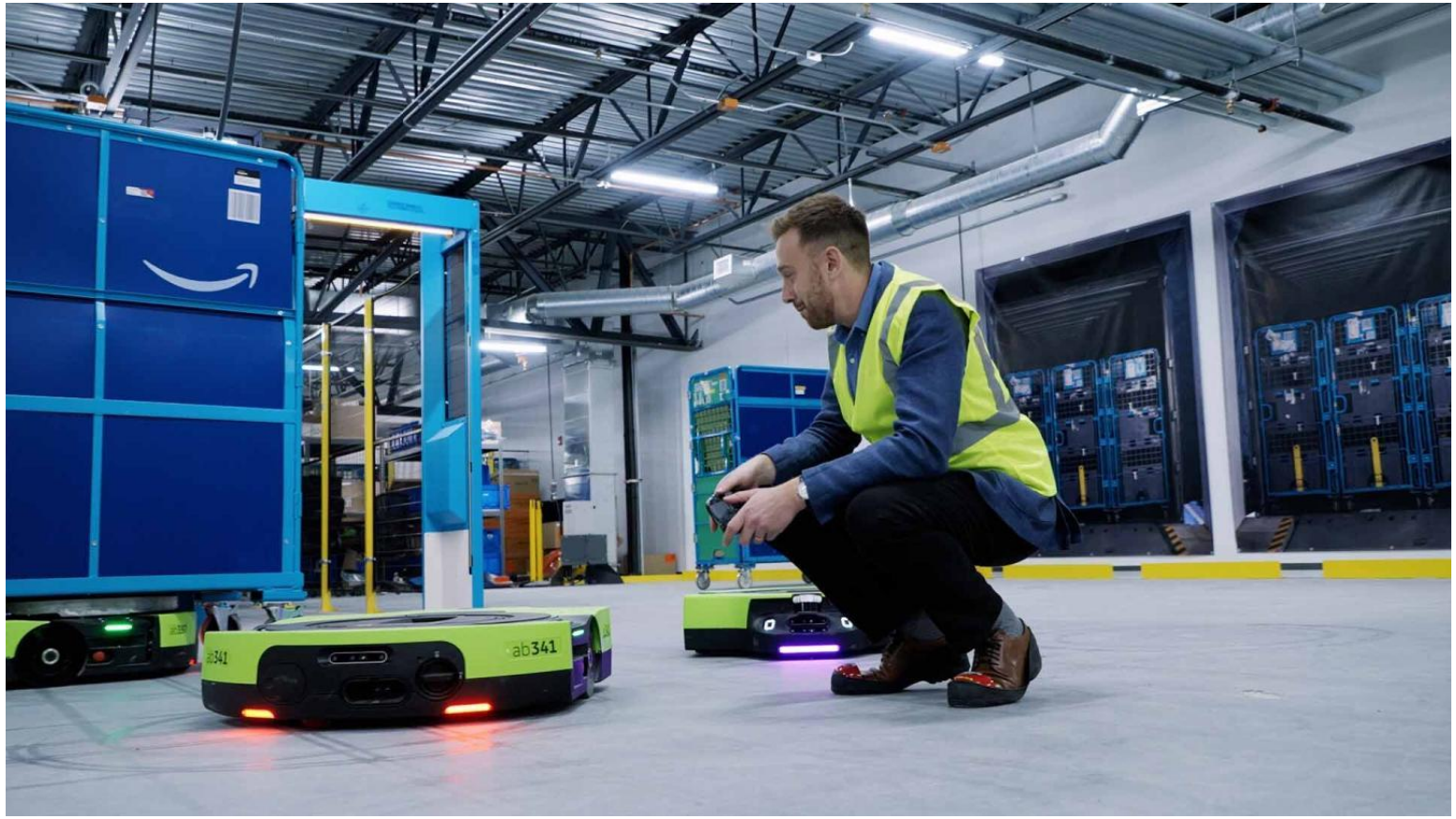


http://www.businessinsider.sg/how-does-googles-waymo-self-driving-car-work-graphic-2017-1/?r=US&IR=T



https://www.tesla.com/robotaxi

Proteus: Amazon's first fully autonomous warehouse robot



https://fortune.com/2025/03/27/amazon-robotics-worker-safety-proteus-warehouse-upskilling/

AI for Health Care

10 AI Applications That Could Change Health Care

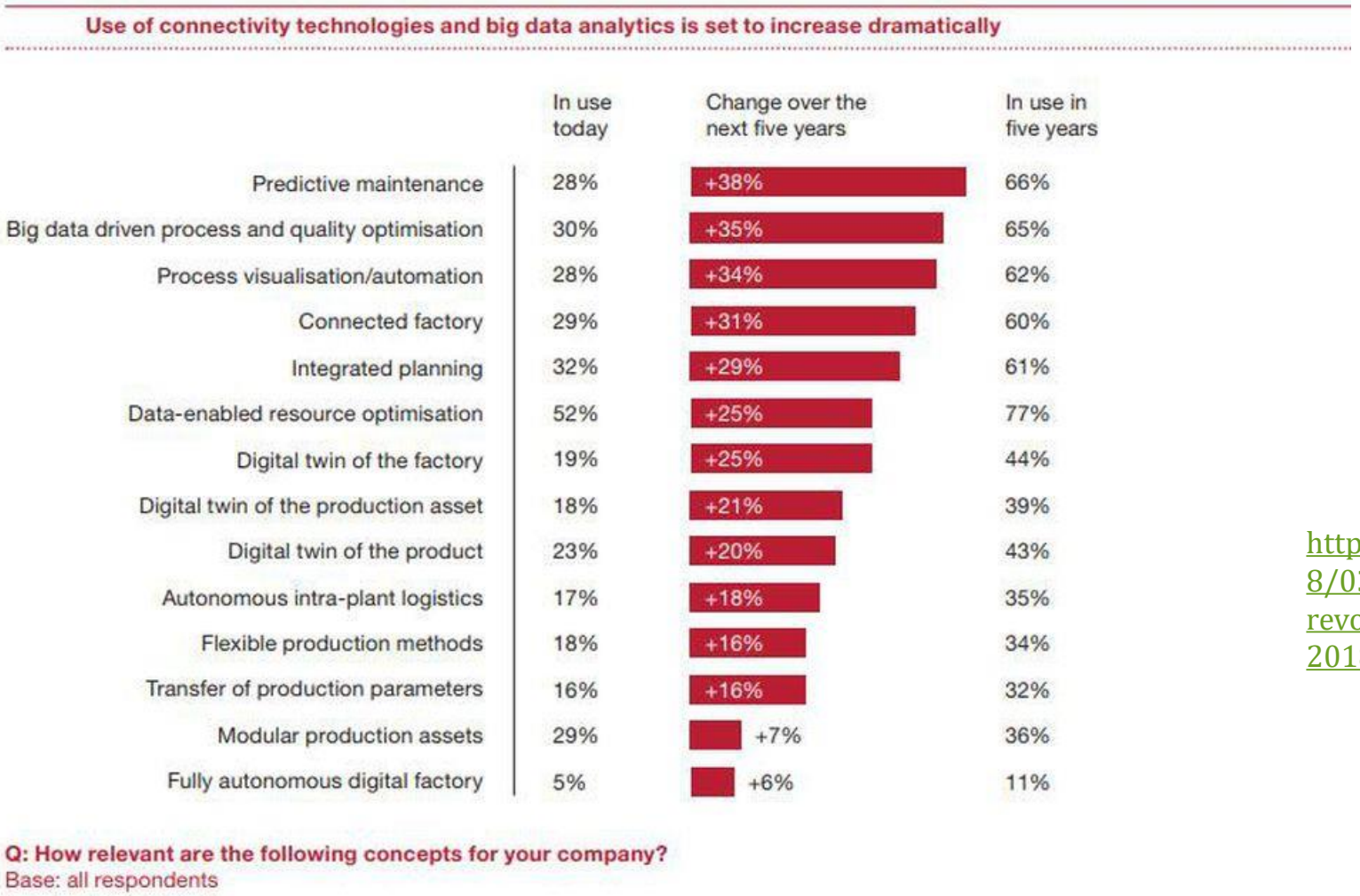
APPLICATION	POTENTIAL ANNUAL VALUE BY 2026	KEY DRIVERS FOR ADOPTION
Robot-assisted surgery	$40B	Technological advances in robotic solutions for more types of surgery
Virtual nursing assistants	20	Increasing pressure caused by medical labor shortage
Administrative workflow	18	Easier integration with existing technology infrastructure
Fraud detection	17	Need to address increasingly complex service and payment fraud attempts
Dosage error reduction	16	Prevalence of medical errors, which leads to tangible penalties
Connected machines	14	Proliferation of connected machines/devices
Clinical trial participation	13	Patent cliff; plethora of data; outcomes-driven approach
Preliminary diagnosis	5	Interoperability/data architecture to enhance accuracy
Automated image diagnosis	3	Storage capacity; greater trust in AI technology
Cybersecurity	2	Increase in breaches; pressure to protect health data

SOURCE ACCENTURE

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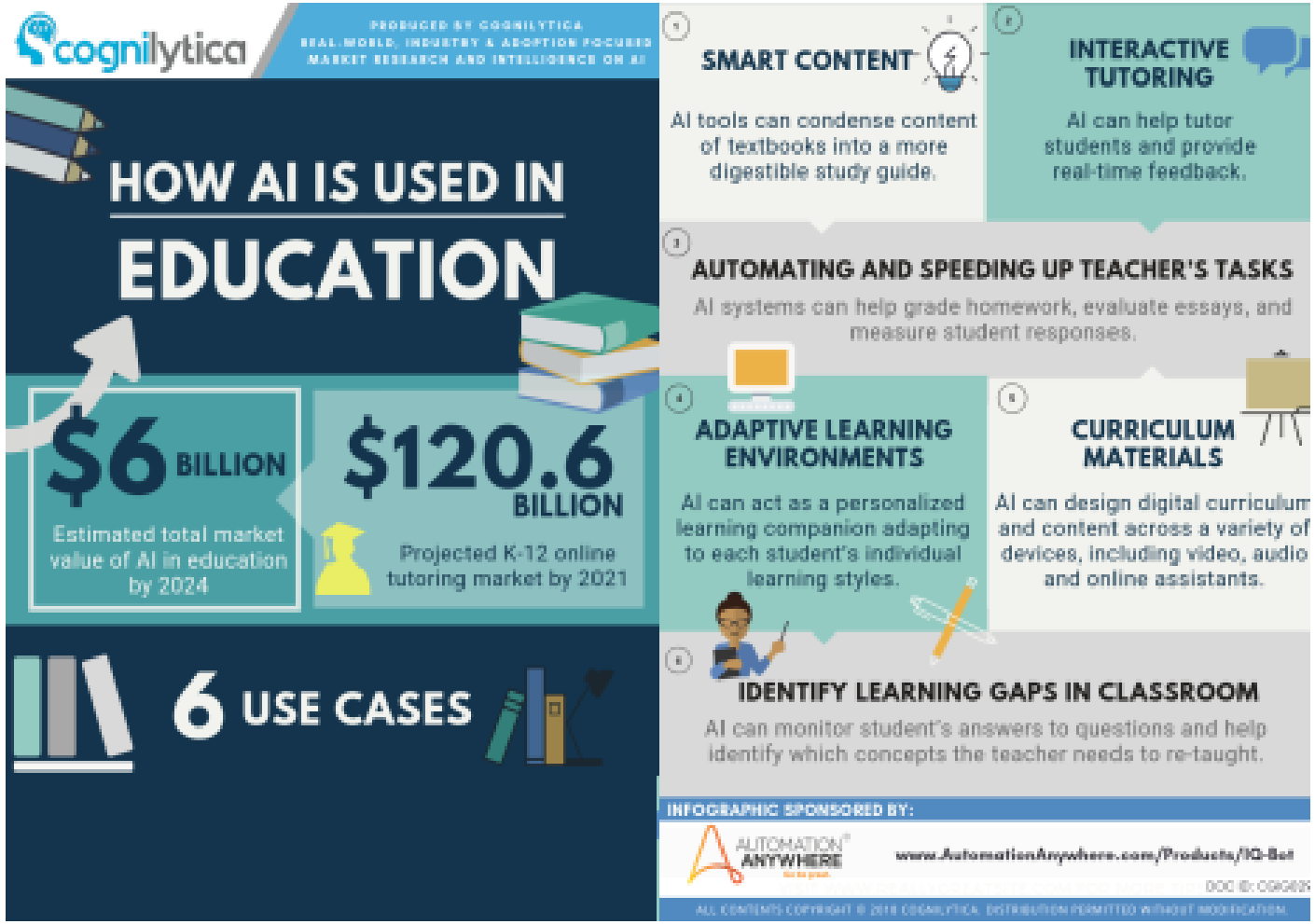
https://hbr.org/2018/05/10-promising-ai-applications-in-health-care

AI for Manufacturing



https://www.forbes.com/sites/louiscolumbus/2018/03/11/10-ways-machine-learning-is-revolutionizing-manufacturing-in-2018/#61e28f2523ac

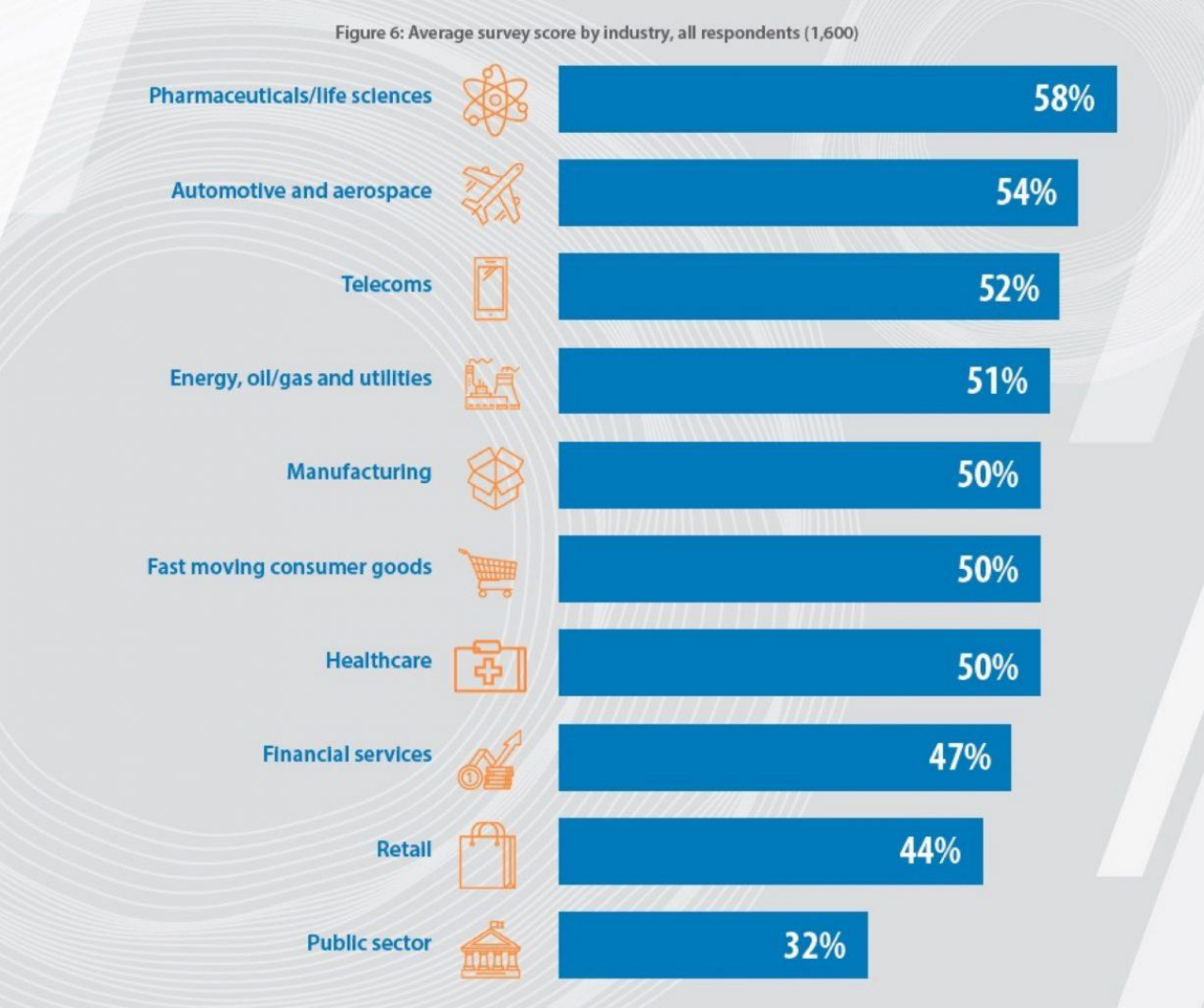
AI for Education



https://www.cognilytica.com/2019/03/28/infographic-ai-in-education/

AI Adoption by Industry

Percentage of market players who plan to adopt AI in next two years across various business verticals (Infosys Survey, 2018)



https://existek.com/blog/ai-programming-and-ai-programming-languages/

Generative AI

- **Generative artificial intelligence** (**generative AI**, **GenAI**, or **GAI**) is a subset of artificial intelligence that uses generative models to produce text, images, videos, or other forms of data. These models often generate output in response to specific prompts.
- Generative AI systems learn the underlying patterns and structures of their training data, enabling them to create new data.
- Examples:
  - Text: GPT-3, GPT-4, GPT-4o, LaMDA, LLaMA, BLOOM, Gemini, Claude
  - Code: Github Copilot, Code T5, OpenAI Codex
  - Image: Imagen, DALL-E 3, Midjourney, Adobe Firefly, FLUX.1
  - Audio and Music: MusicLM and MusicGen



Théâtre D'opéra Spatial, an image made using generative artificial intelligence

AI generated images have become much more advanced.





AI Dangers

- **Bias and Inaccuracies**: AI systems may reinforce harmful biases and misinformation if their training data is flawed or biased.
- **Privacy Concerns**: Extensive data collection by AI platforms can compromise personal privacy and lead to misuse of sensitive information.
- **Loss of Human Judgment**: Over-reliance on AI might diminish our ability to make independent decisions and critically evaluate outcomes.
- **Deepfakes and Manipulation**: AI can create convincing fake content that threatens truth, trust, and societal stability.
- **Loss of Human Control**: Automation of critical decisions might reduce human oversight, creating potential for serious unintended consequences.
- **Employment Disruption**: AI-driven automation could displace workers, exacerbating economic inequalities and social tensions.
- **Existential and Long-term Risks**: Future advanced AI, such as AGI (Artificial General Intelligence), could become misaligned with human interests, posing significant existential threats.

https://www.jdsupra.com/legalnews/afraid-of-ai-learn-the-seven-cardinal-2026474/

OECD.AI Policy Observatory | GPAI Global Partnership

OECD | Going Digital | DPP

Blog | Live data | Policies and initiatives | Priority issues | Tools | Resources | About

Home > AIM > AI-Generated Fake Medical Research by Indonesians Sparks Academic Scandal

AI-Generated Fake Medical Research by Indonesians Sparks Academic Scandal



AI principles

Accountability | Transparency & explainability

Industries

Education and training | Healthcare, drugs, and biotechnology

Affected stakeholders

General public | Civil society

Harm types

Reputational | Public interest

AI system task:

Content generation

https://oecd.ai/en/incidents/2026-05-26-424e

2026 onward: Agentic AI

- Agentic AI refers to autonomous systems that do not just generate text or answer questions, but independently perceive their environment, plan multi-step workflows, use tools or APIs, execute actions, and reflect on the results. (https://cloud.google.com/discover/what-is-agentic-ai)
- Real world examples (https://www.simpliaxis.com/resources/agentic-ai-examples)
  - **PayPal:** Uses autonomous agents for real-time fraud detection and transaction processing
  - **Oxford University Hospital**: Working with Microsoft, the hospital built agentic systems into clinical workflow support, coordinating administrative steps around patient care while leaving clinical judgment entirely to humans.
  - **Siemens**: Siemens introduced its Fuse EDA AI Agent in March 2026, a domain-scoped autonomous agent that plans and orchestrates multi-tool workflows across the entire design-to-sign-off pipeline.

Discussions

- Provide examples of how AI technology is used within the fields of study relevant to your faculty or school.
- Discuss the impact of using such technology across various dimensions, for instance:
  - Does it reduce the need for human labor?
  - What is its impact on the environment?
  - Does it lead to greater efficiency in systems or processes?
  - …
- How can Generative AI be utilized ethically within an academic context?

Have fun studying!